

Problem Solving in Engineering (IP-3)

Module 1 Workbook: Serenity VR

Select a Super System

Healthcare and Mental Wellness System

Select a Specific Sub System (Problem Area)

- **Sub System:** Anxiety Disorder Management and Therapy Delivery System
- **Specific Location:** Urban India (Hyderabad / Metropolitan cities)
- **Specific Issue:** Inaccessible, non-adaptive, and delayed anxiety treatment
- **Specific Focus:** Patients with anxiety disorders who lack access to timely, personalized, biofeedback-driven therapy

Problem Area

India has approximately 38 million people living with anxiety disorders, yet 85–90% do not receive adequate treatment due to a severe shortage of mental health professionals (only ~0.3 psychiatrists per 100,000 population vs. the WHO-recommended 3 per 100,000). Existing therapy methods are expensive, require frequent in-person clinic visits, offer no real-time physiological feedback, and cannot dynamically adapt to a patient's momentary stress levels. This results in delayed intervention, poor treatment adherence, and compounding anxiety - particularly among urban youth and working professionals who face high-stress environments but lack affordable, technology-driven support systems.

Study the System (PID - 5W2H Framework)

WHAT is/are the SYMPTOM(s) (Effect) of the problem observed?

1. **Massive treatment gap:** 85–90% of individuals with mental health conditions in India do not receive the care they need [Source: Mental Health Foundation of India (mh-faindia.com); PIB, Govt. of India].
2. **Rising anxiety prevalence:** Anxiety disorders in India have increased by over 120% between 1990 and 2023 [Source: Centre for Mental Health Law and Policy (CMHLP); Global Burden of Disease Study].
3. **Non-adaptive therapy:** Current treatment protocols (CBT, medication) follow fixed schedules and do not adapt in real-time to a patient's physiological state during sessions.
4. **Post-pandemic surge:** COVID-19 caused a marked rise in anxiety and depression in both primary care and general population settings across India [Source: National Medical Journal of India (NMJI)].

5. **High dropout rates:** Patients discontinue therapy due to cost, travel burden, stigma, and lack of perceivable real-time progress.

WHO is affected most by the problem? (Stakeholder Group 1)

1. **Patients with anxiety disorders** - Particularly urban youth (18–35), college students, and working professionals who face academic pressure, workplace stress, and social media-driven anxiety.
2. **Women and adolescents** - Identified as groups experiencing a disproportionately higher burden of anxiety and depressive disorders [Source: CMHLP].
3. **Patients in rural/semi-urban areas** - Over 60% of India's population resides in rural areas where mental health professionals are virtually absent [Source: psychology.town; lyfsmile.com].

WHO is Causing the Problem? (Stakeholder Group 2)

1. **Systemic shortage of mental health professionals** - India has ~0.3–0.75 psychiatrists per 100,000 population against the WHO recommendation of 3 per 100,000 [Source: PIB, Govt. of India; mentis.co.in].
2. **Healthcare policy gaps** - Insufficient funding and infrastructure for mental health under the National Mental Health Programme.
3. **Social stigma** - Cultural attitudes that discourage seeking mental health treatment, leading to delayed diagnosis and care.
4. **Technology gap in therapy delivery** - Absence of affordable, technology-assisted, real-time adaptive therapy systems.

WHEN (Time patterns) is the Symptom happening?

- **Chronic and ongoing** - Anxiety disorders are persistent conditions that worsen without intervention.
- **Peak during high-stress periods** - Exam seasons (March–May), work deadlines, and financial year transitions.
- **Post-pandemic acceleration** - Significant worsening since 2020, with no sign of reversal.
- **Daily triggers** - Morning commutes, workplace hours (9 AM – 7 PM), and late-night screen time are common trigger periods.
- **When symptoms are NOT occurring** - During restful sleep, vacation periods, or when patients are actively engaged in therapeutic activities (meditation, exercise, nature exposure).

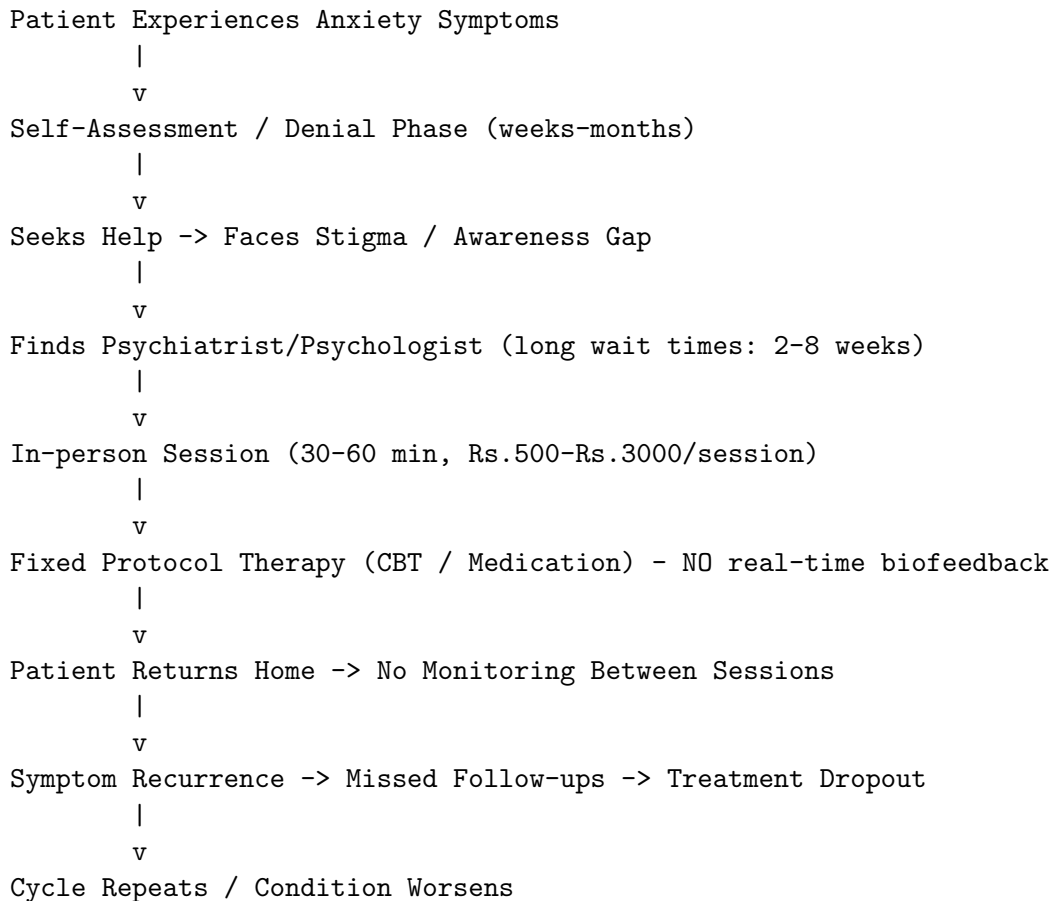
WHERE (Space context) is/are the symptoms observed?

- **Urban metropolitan areas** - Hyderabad, Bangalore, Delhi, Mumbai - where high-stress lifestyles are prevalent.
- **Educational institutions** - Engineering colleges, medical schools, competitive exam coaching centres.
- **Workplaces** - IT corridors, startup ecosystems, BPO/call centres.

- **NOT prominently observed** - In communities with strong social support networks, active wellness programs, or access to regular counselling.

HOW is the process cycle observed?

Current Therapy Delivery Process Flow:



Key Observation: The current system is *reactive* (wait for crisis), *non-adaptive* (same protocol regardless of real-time state), and *disconnected* (no data between sessions).

HOW MUCH is the impact?

Write the Problem Statement (Preliminary)

Approximately 38 million Indians suffer from anxiety disorders, yet 85–90% remain untreated due to a critical shortage of mental health professionals (~0.3 psychiatrists per 100,000 vs. WHO's recommended 3 per 100,000), high therapy costs, social stigma, and the absence of real-time adaptive intervention systems. Current therapy is episodic, non-responsive to real-time physiological states, and disconnected between sessions - leading to poor adherence, delayed intervention, and a compounding feedback loop where untreated anxiety worsens over time, further discouraging patients from seeking help.

Are these problems related to each other?

Yes - the shortage of professionals drives up costs and wait times → patients delay seeking help → anxiety compounds → when they finally seek help, the condition is more severe → fixed-protocol therapy is less effective for severe cases → patients drop out → the treatment gap widens further.

Metric	Data Source
People with anxiety disorders in India	~38 million Mental Health Foundation of India
Treatment gap	85-90% do not receive care PIB, Govt. of India
Increase in anxiety disorders (1990-2023)	Over 120% CMHLP / Global Burden of Disease
Psychiatrists per 100,000 population	~0.3 vs. 3.0 (WHO) PIB; WHO
Clinical psychologists per 100,000	~0.07 tyfsmile.com; psychology.town
Average therapy cost per session	Rs.500-3000 Market data, urban India
Population in India with mental health disorder	~11 million in 100 (over 150 million) PIB, Govt. of India

What is the Causal relationship?

Professional shortage → high cost and wait times → delayed treatment → worsened condition → poor therapy outcomes → dropout → untreated population grows → increased demand on already insufficient supply.

Which one problem is the root cause?

The **absence of affordable, scalable, real-time adaptive therapy delivery systems** is the core enabler of the vicious cycle. While the professional shortage is structural, a technology-assisted therapy system can bridge the gap by providing accessible, biofeedback-driven, AI-personalized anxiety management.

Is there a feedback loop?

Yes - a reinforcing (vicious) loop:

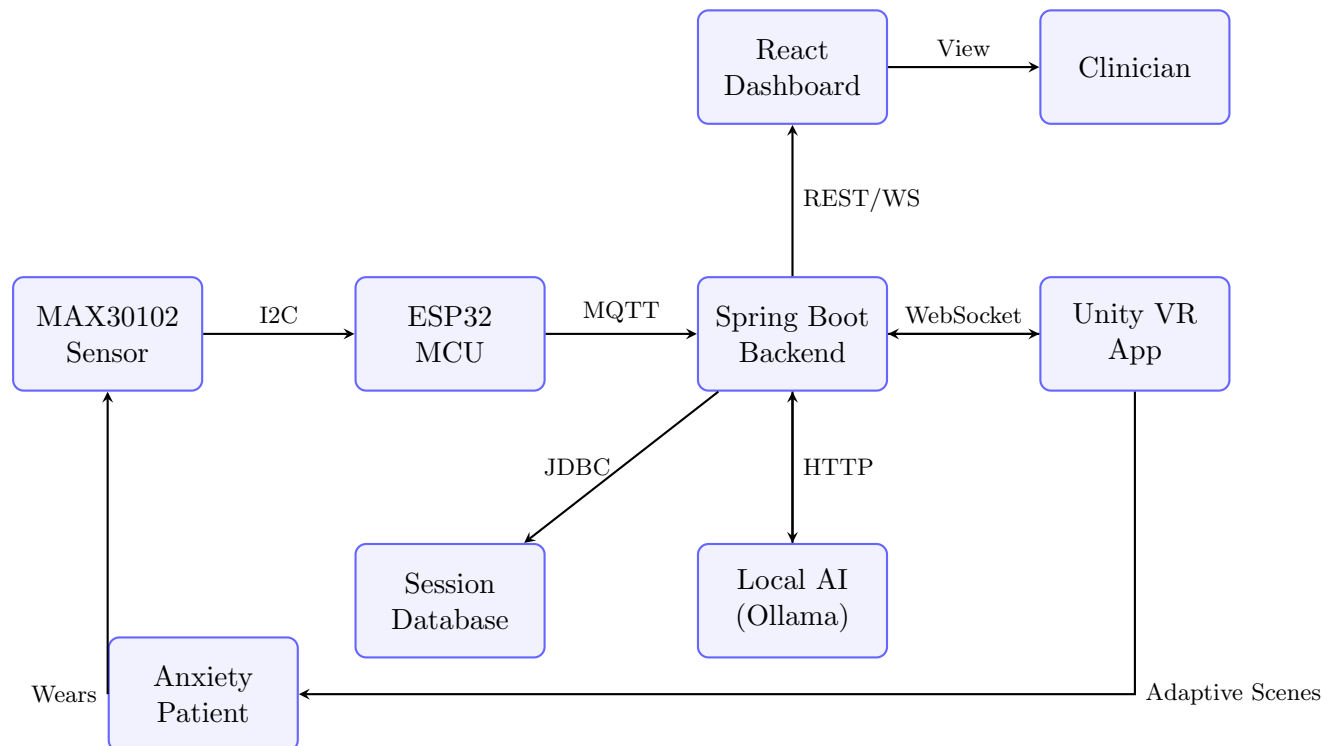
Untreated anxiety → Increased Severity → Higher Cost of treatment → Increased Patient avoidance → Larger Untreated population → Higher Demand on limited professionals → Longer Wait times → More Untreated anxiety (loop continues).

Define the System

System Components (defined by function)

1. **Anxiety Patient:** The end-user experiencing anxiety symptoms, seeking therapy
2. **Biometric Sensor (MAX30102 + ESP32):** Measures real-time SpO2 and pulse rate of the patient
3. **VR Headset (Meta Quest 3):** Delivers immersive therapeutic environments (guided breathing, nature scenes)
4. **Unity VR Application:** Renders adaptive VR scenes that respond to biometric data
5. **Spring Boot Backend:** Central hub - routes biometric data, AI responses, and session data
6. **Local AI Model (Ollama - Llama 3 8B):** Generates personalized therapy guidance based on biometric trends
7. **React Dashboard:** Displays live biometric data, session history, and analytics
8. **Clinician / Therapist:** Monitors patient data via dashboard, adjusts treatment parameters
9. **MQTT / WebSocket Protocol:** Real-time data transport layer connecting all components
10. **Session Database:** Stores historical biometric data, session logs, and patient progress

System Map



Define the System Boundary (Scope)

What is **INSIDE** the system boundary (scope of study):

- Biometric sensor hardware (MAX30102 + ESP32)
- VR headset and Unity application
- Spring Boot backend server
- Local AI inference container (Ollama)
- React monitoring dashboard
- Real-time data pipeline (MQTT/WebSocket)
- Session data storage

What is **OUTSIDE** the system boundary (environment):

- Pharmaceutical treatment / medication
- In-person clinical therapy sessions
- Hospital/clinic infrastructure
- Health insurance systems
- Government mental health policy
- Patient's social/family support system
- Internet Service Provider / Network infrastructure

Inputs (from environment INTO the system):

- Patient's physiological signals (heartbeat, SpO2) [GREEN ARROW]
- Patient's willingness to engage with VR therapy [GREEN ARROW]
- Clinician's treatment parameters and preferences [GREEN ARROW]
- Pre-trained AI model weights (Llama 3) [GREEN ARROW]
- Power supply and network connectivity [GREEN ARROW]

Outputs (from the system TO the environment):

- **Desirable:** Reduced anxiety levels, improved patient self-awareness, therapy session reports for clinicians, historical trend data [RED ARROW]
- **Undesirable (waste):** Heat generation from GPU/hardware, potential sensor inaccuracy/noise, VR-induced motion sickness risk, data privacy concerns [RED ARROW]

System Boundary Diagram



Stakeholder Map

1. Stakeholder: Anxiety Patients (End Users)

Needs:

- Affordable, accessible anxiety management tool that does not require frequent clinic visits
- Real-time feedback on their physiological state during therapy (biofeedback)
- Immersive, calming VR environments that adapt to their stress levels
- Privacy and data security for sensitive health information
- Ability to track progress over time through session history

2. Stakeholder: Clinicians / Therapists

Needs:

- Real-time dashboard to monitor patient biometrics (SpO2, pulse) during and between sessions
- Historical session data and trends to inform treatment decisions
- Ability to customize VR therapy parameters (scene type, breathing pace, session duration)
- Alerts for abnormal biometric readings
- Integration with existing clinical workflows

3. Stakeholder: Healthcare Institutions / Hospitals

Needs:

- Scalable solution that extends their mental health service capacity without proportional increase in staff
- Cost-effective alternative to fully in-person therapy delivery
- Evidence-based outcomes data to justify investment
- Compliance with data privacy standards (e.g., IT Act 2000, upcoming DPDP Act)
- Interoperability with existing Electronic Health Record (EHR) systems

4. Stakeholder (Additional): Caregivers / Family Members

Needs:

- Visibility into patient's therapy progress (with patient consent)
- Assurance that the system is safe and medically guided
- Simple, non-technical interface for understanding patient status

Quantitative Baseline Data

1. Data Point A:

Approximately 38 million people in India live with anxiety disorders, and the prevalence has increased by over 120% between 1990 and 2023. Despite this, 85–90% of individuals with mental health conditions do not receive adequate treatment, constituting one of the largest treatment gaps globally.

Source: Centre for Mental Health Law and Policy (CMHLP), Global Burden of Disease Study; Mental Health Foundation of India (mhfaindia.com); Press Information Bureau (PIB), Government of India.

2. Data Point B:

India has only ~0.3–0.75 psychiatrists per 100,000 population and ~0.07 clinical psychologists per 100,000 population, compared to the WHO recommendation of at least 3 psychiatrists per 100,000. This means India would need approximately **42,000+ psychiatrists** to meet the WHO standard, but currently has fewer than **10,000**.

Source: Press Information Bureau (PIB), Government of India; WHO Mental Health Atlas; mentis.co.in; psychology.town.

3. Data Point C (Additional):

Meta-analyses show that VR-based exposure therapy (VRET) is as efficacious as traditional in-vivo exposure therapy for anxiety disorders, with significant reductions in self-reported anxiety and physiological markers (heart rate, HRV). VR-based biofeedback systems have demonstrated measurable improvements in heart rate variability and subjective anxiety scores.

Source: Frontiers in Psychology (frontiersin.org); National Institutes of Health (NIH) - PubMed Central; Journal of Medical Internet Research (JMIR).

State "How Might We..."

1. How Might We

How might we move from a **therapy system that requires patients to physically visit scarce mental health professionals** on fixed schedules to a system that **delivers accessible, VR-based anxiety therapy at home**, using real-time biometric feedback to adapt the therapeutic experience to the patient's momentary physiological state?

2. How Might We

How might we move from **static, one-size-fits-all therapy protocols** that do not account for real-time patient physiology to a system where **a local AI model dynamically personalizes therapeutic guidance** (breathing exercises, scene selection, pacing) based on live SpO2 and pulse data from wearable sensors?

3. How Might We

How might we move from **disconnected, episodic therapy with no data continuity between sessions** to a system that **continuously tracks biometric trends, provides clinicians with live dashboards and historical analytics**, and enables data-driven treatment decisions - thereby improving treatment adherence and reducing the 85–90% treatment gap?